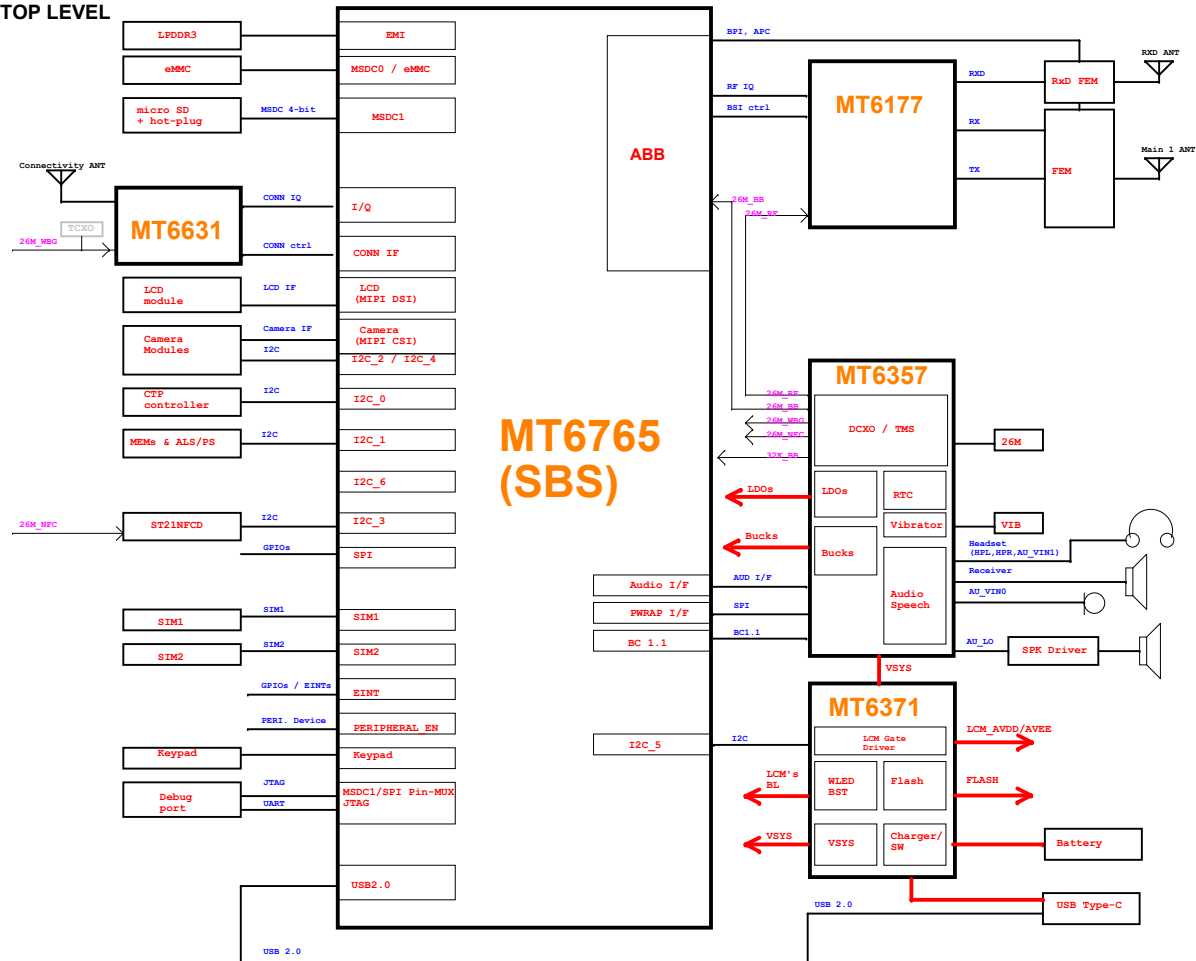


Project : MT6765 LPDDR3
REF_SCH TOP LEVEL



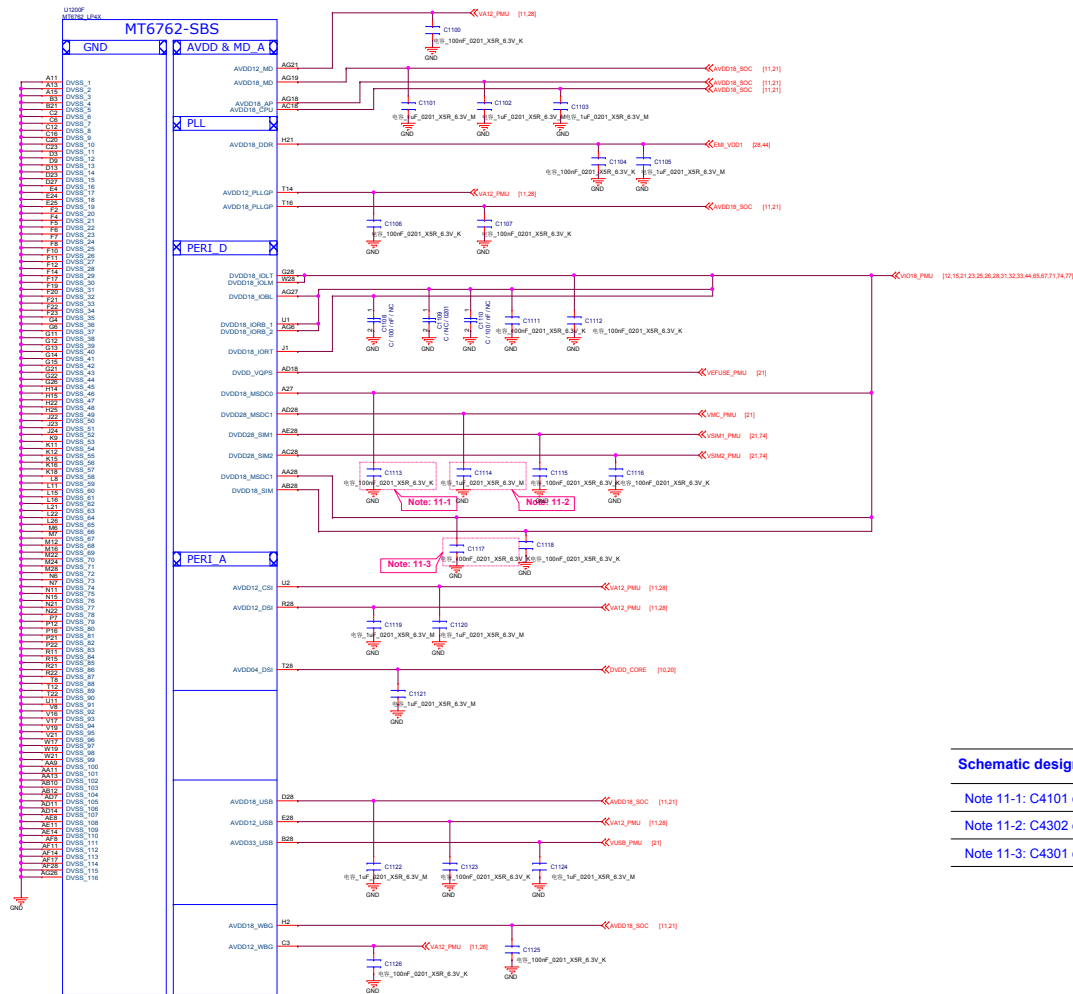
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 01_Block_Diagram		
Size D	Project F9_MB_V1	Rev V1
Date: Friday, October 19, 2018	Sheet 1 of 99	

I2C	Sub SYS	Function	Part Number	I2C Spec.	i2C Slave Address / Write / Read (7-bit mode)	
I2C-0	AP	Cap Touch controller	GT1151	400 Kbps	0x5D	Write:0xBA / Read:0xBB
I2C-1 (I3C)	AP Sensor Hub	Magnetic Sensor	AK09918C	400 Kbps	0x0C	Write:0x18 / Read:0x19
		Ambient Light Sensor	CM36558	400 Kbps	0x51	Write:0xA2 / Read:0xA3
		Proximity Sensor				
		Pressure Sensor	BMP280	400 Kbps	0x77	Write:0xEE / Read:0xEF
I2C-2 (I3C)	AP	Rear Camera	IMX230	400 Kbps	0x1A	Write:0x34 / Read:0x35
			EEPROM	400 Kbps	0x50	Write:0xA0 / Read:0xA1
			AF driver	400 Kbps	0x0C	Write:0x18 / Read:0x19
I2C-3	AP	Audio Smart PA	RT5510	400 Kbps	0x34	Write:0x68 / Read:0x69
		NFC	ST21NFCD	400 Kbps	0x08	Write:0x10 / Read:0x11
I2C-4 (I3C)	AP	Front Camera	S5K217	400 Kbps	0x2D	Write:0x5A / Read:0x5B
			EEPROM	400 Kbps	0x52	Write:0xA4 / Read:0xA5
I2C-5	AP	Sub-PMIC	MT6371 PMU	3.4 Mbps	0x34	Write:0x68 / Read:0x69
			MT6371 PD	3.4 Mbps	0x4E	Write:0x9C / Read:0x9D
I2C-6	AP					
Note : I2C Spec. : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)						

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 02_I2C_ID_Overview		
Size C	Project F9_MB_V1	Rev V1
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Date	Category	Item
2017.11.24 (V0.1)	Page 05	V0.1 Release
2017.12.7 (V0.2)	Page 11	Change power of AVDD18_DDR(H21) from VIO18_PMU to EMI_VDD1, connecting EMI_VDD1 to SH2102 in star connection
	Page 12	Add Note 12-5
	Page 21	Add SH2102 for star connection among EMI_VDD1, AVDD18_SOC, and VIO18_PMU
	Page 22	1. Change C2304 from C / 1 / uF / 10V to C / 1 / uF / 6.3V 2. Change R2301 from R / 1.5 / K to R / 7.5 / K 3. Update Note 22-5
	Page 44	1. To enhance LPDDR3 power off sequence,change C4422 from 0.1uF to 2.2uF, and change C4423 from 0.1uF to 1uF 2. Change VDD1 power of eMCP from VIO18_PMU to EMI_VDD1, fulfilling power rail in star connection
	Page 51	1. Add 2nd source plan for U5004 2. Add Note 51-3 and Note 51-4

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 05_Change_Notice		
Size C	Project F9_MB_V1	Rev V1
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Schematic design notice of "11_BB_POWER_IO" page.

Note 11-1: C4101 closed DVDD18_MSDC0 150mil

Note 11-2: C4302 closed DVDD28_MSDC1 150mil

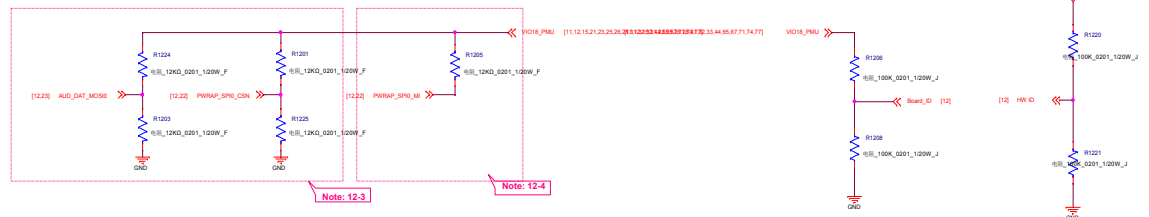
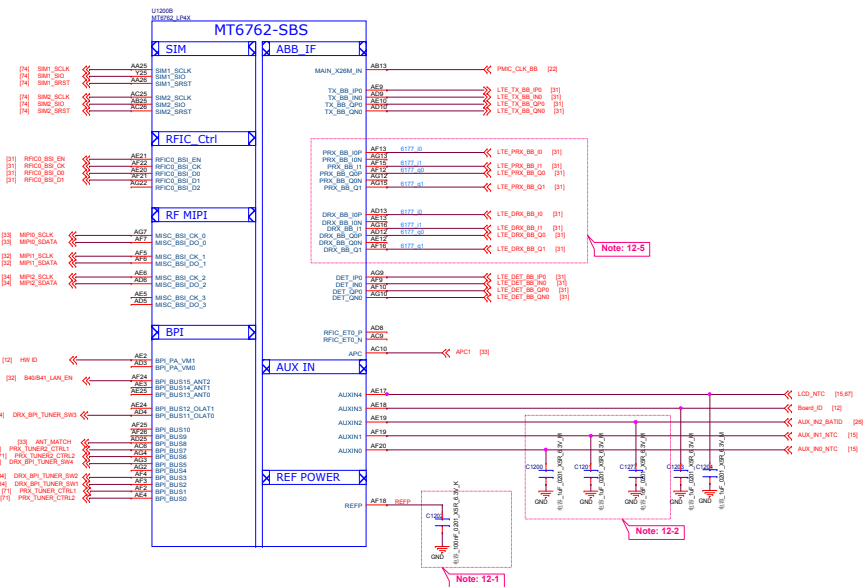
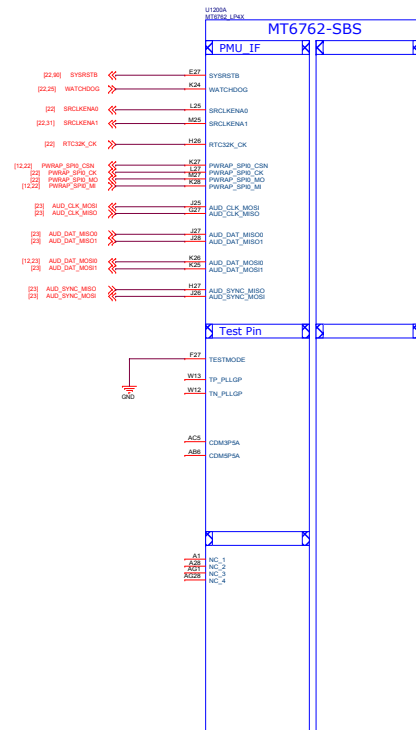
Note 11-3: C4301 closed DVDD18_MSDC1 150mil

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Title 11_BB_POWER_IO

Size D	Project F9_MB_V1	Rev v1
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Schematic design notice of "12_BB_1" page.

Note 12-1: The de-coupling cap. for REFP (AF18 ball) have to be placed as close to BB as possible.

Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-3: "PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pin to select which interface will be the JTAG pin out.

PWRAP_SPI0_CSN	AUD_DAT_MOSI0	AP_JTAG	MT_JTAG
default=PU	default=PD	N/A	N/A
HI	LO	N/A	N/A
HI	HI	SPI0+EINT8	SPI1+SPI3
LO (by ext. PDI)	LO	SPI0+EINT8	N/A
LO (by ext. PDI)	HI	MSDC1	N/A

Note 12-4: PWRAP_SPI0_M and PWRAP_SPI0_Mi are DDR type feature in bootstrap

PWRAP_SPI0_Mi	Booting interface	MSDC0 pin mux
default=PU	DDR	follow LP3 Ref SCH.
LO (by ext. PDI)	LPDDR3	follow LP3 Ref SCH.
HI	LPDDR4X	follow LP4X Ref SCH.

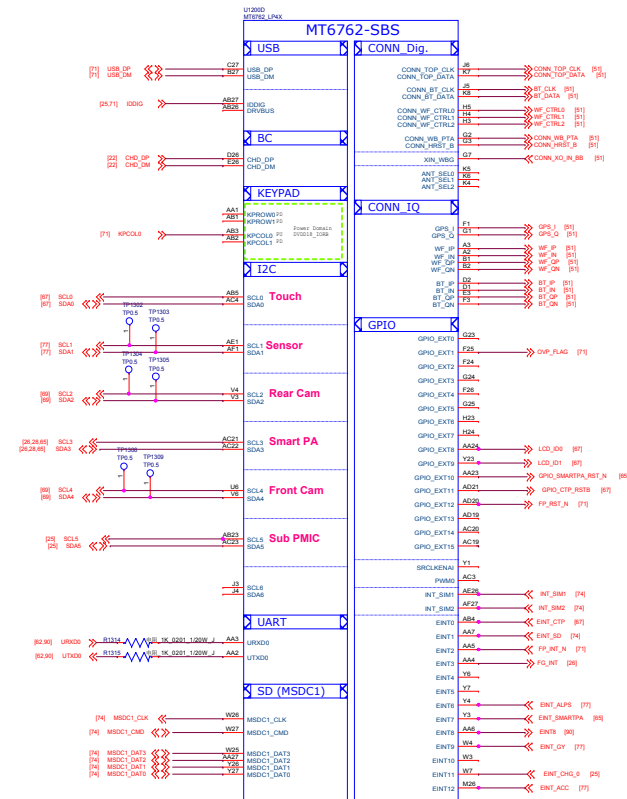
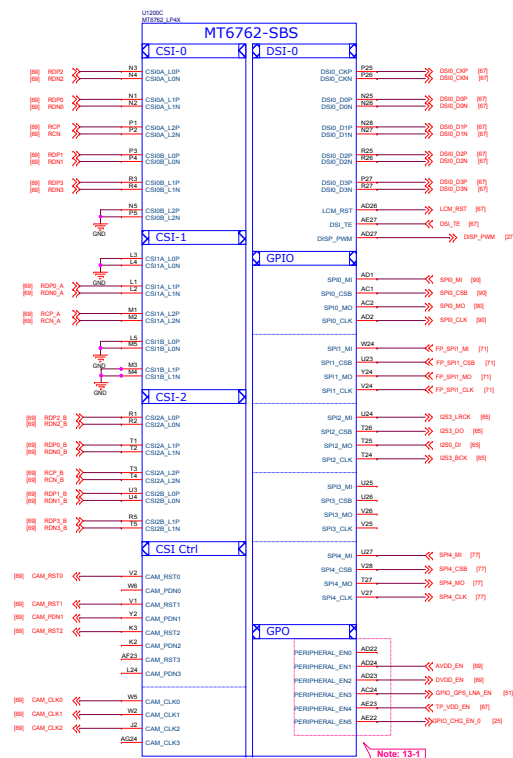
Note 12-5: Please set unused IQ pins in NC

华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 12_BB_1

Size D Project F9_MB_V1 Rev v1

Date: Friday, October 19, 2018 Sheet 12 of 99



Schematic design notice of "13_BB_2" page.

Note 13-1: The enable pin of acoustic or optoelectronic devices (e.g. SPK AMP/Backlight/Charger OCP/OVP) suggest to use Peripheral_EN[0:5].
If use other GPIOs as enable pin, suggest to reserve 0201 NC to GND

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Title 13_BB_2

Size D Project F9_MB_V1

Rev V1

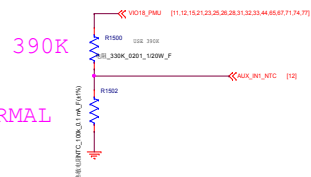
Date: Friday, October 19, 2018

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LCD_THERMAL



PA_THERMAL

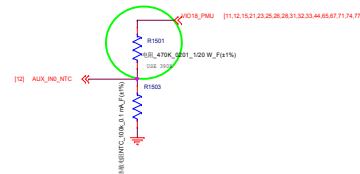


change to NCP03WF104F05RL Murata

Thermistor to sense RF PA temperature

1. NTC1502 must close to LTE Band 7 PA or the hottest PA <2mm.
2. The distance is the shortest distance from package edge to edge.

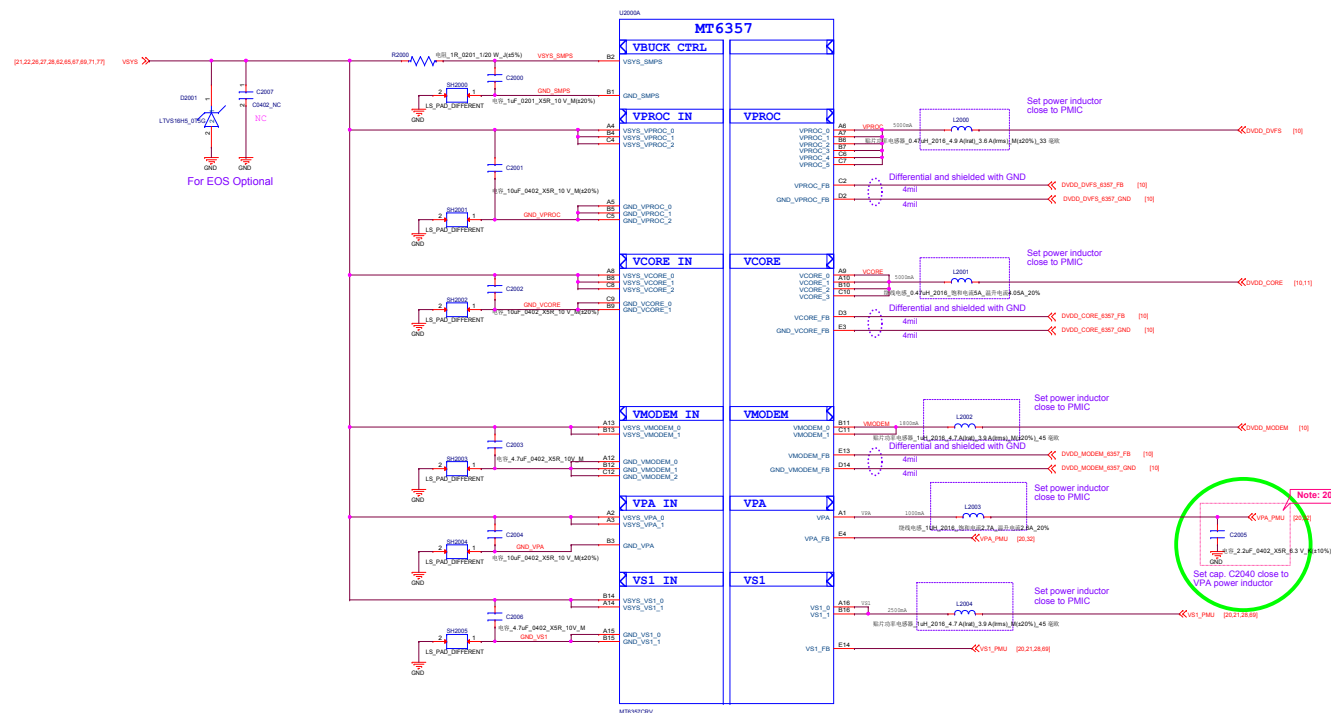
Thermistor / To sense board level temperature



Thermistor to sense AP temperature

1. NTC1501 must keep a distance about 6-8 mm away from AP and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

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Title 15_BB_AUXADC_Thermal		
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Schematic design notice of "20_POWER_MT6357_Buck"

Note 20-1: C2040, please choose 0402 size

Note 20-2: PMIC Part number notice for MT6765/62/61 platform

MTK Platform	PMIC
MT6765 / 62	MT6357 CRV
MT6761	MT6357 MRV

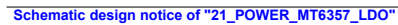
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 20_POWER_MT6357_Buck

Size D	Project F9_MB_V1	Rev V1
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Input Power	Power Name	Output Voltage (V)	Output Current	Default Voltage
LDO from VBAT	VFE28	2.8	40mA	2.8V
	VXO22	2.24	25mA	2.24V
	VCN28	2.8	40mA	2.8V
	VCAMA	1.8/2.5/2.8	200mA	2.8V
	VAUX18	1.8	20mA	1.8V
	VAUD28	2.8	50mA	2.8V
	VBIF28	2.8	1mA	2.8V
	VCN33	3.3/3.4/3.5/3.6	800mA	3.3V
	VLD028	2.8/3.0	360mA	2.8V
	VIO28	2.8	200mA	2.8V
	VMC	1.86/2.9/3.0/3.3	200mA	3.0V
	VMCH	2.9/3.0/3.3	800mA	3.0V
	VEMC	2.9/3.0/3.3	800mA	3.3V
	VS1M1	1.7/1.8/1.86/2.76/3.0/3.3	200mA	1.86V
	VS1M2	1.7/1.8/1.86/2.76/3.0/3.3	200mA	1.86V
LDO from VS1	VIBR	1.2/1.3/1.5/1.8/2.0/2.8/3.0/3.3	200mA	2.8V
	VUSB	3.07	50mA	3.07V
	VRF18	1.81	450mA	1.81V
	VMIFI	1.71/1.8/1.84	300mA	3.0V
	VCN18	1.8	200mA	1.8V
	VCAMD	1/1.05/1.1/1.2/1.3/1.5/1.8	600mA	1.2V
LDO from VS2	VCAMIO	1.8	300mA	1.8V
	VIO18	1.8	1200mA	1.8V
	VRF12	1.2	200mA	1.2V
	VA12	1.2	300mA	1.2V
	VSRAM_PROC	0.6-1.31	400mA	1.1V
	VSRAM_OTHERS	0.6-1.31	200mA	0.9V
	VSRAM_GPU	0.6-1.31	400mA	1.0V
	VDRAM	1.12/1.24	1000mA	1.24V

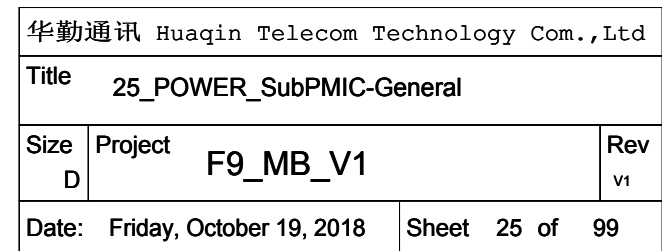


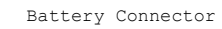
Note 21-3: Ext Buck BOM option

	Ext. buck option	
	w/ EXT VS2 Buck	w/o EXT VS2 Buck
C2104	10uF	22uF
R2851	0-ohm , 0603	NC
R2852	NC	0-ohm , 0603

Please also refer to MT6357 design notice for further detail design information

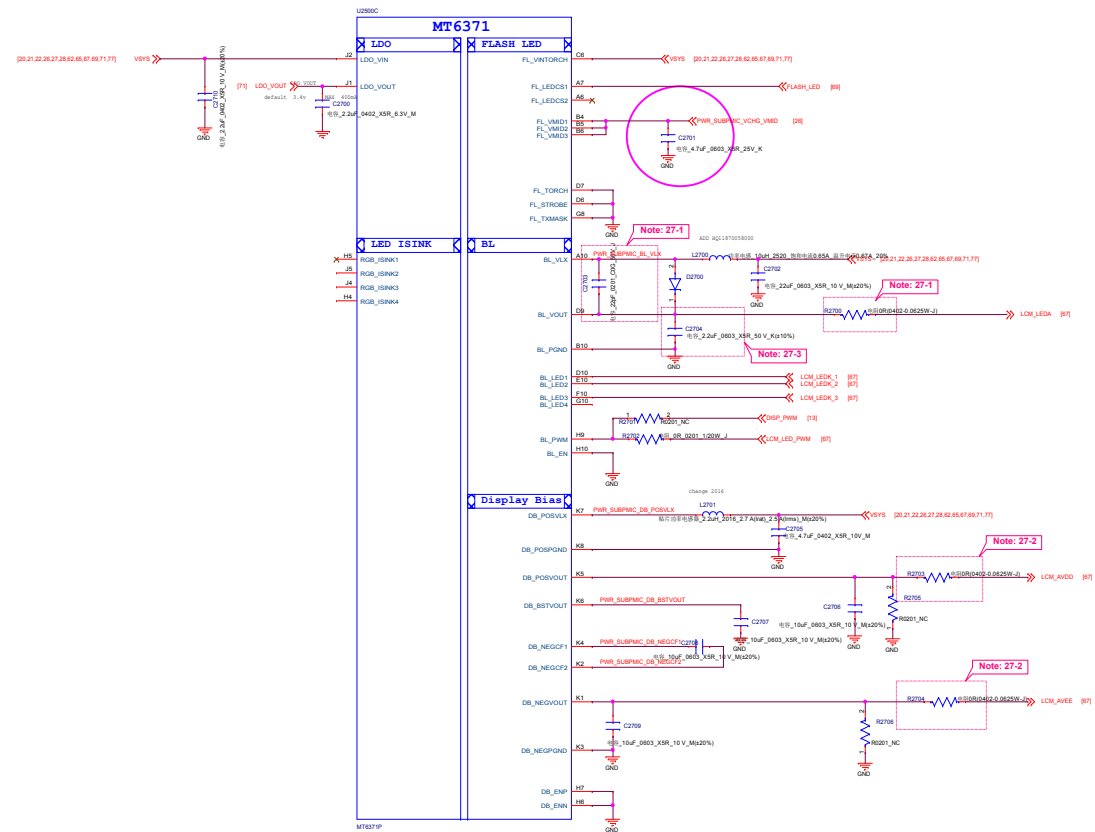
Note 21-5: Please connect VS2_LDO1(F15) to VS1_PMU if voltage applied to VCAMD(E17) ≥ 1.3 V



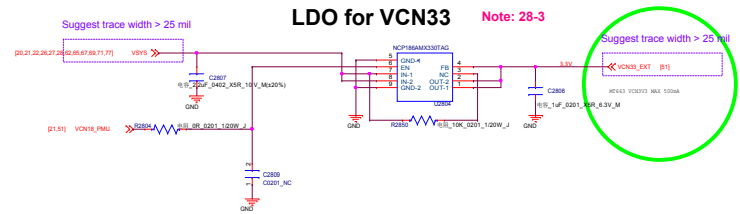
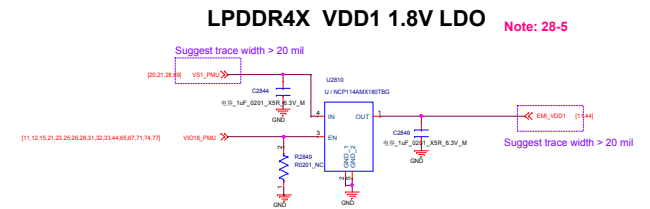
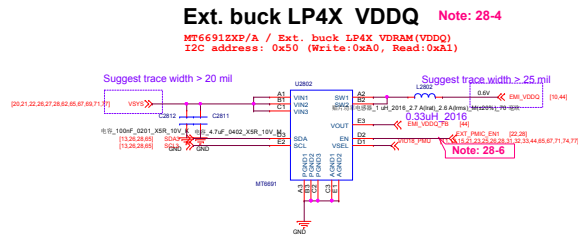
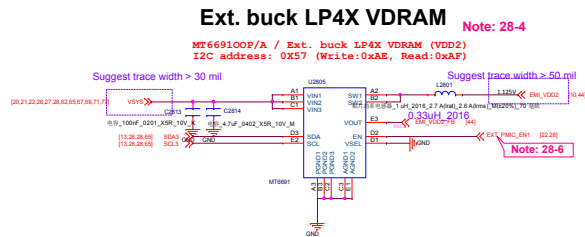
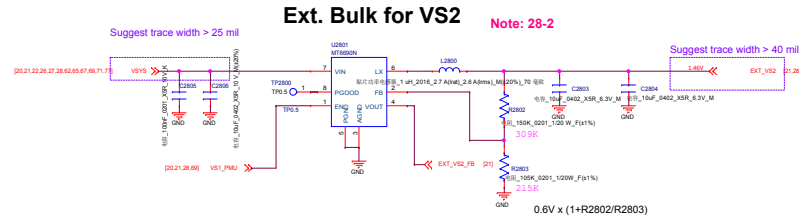
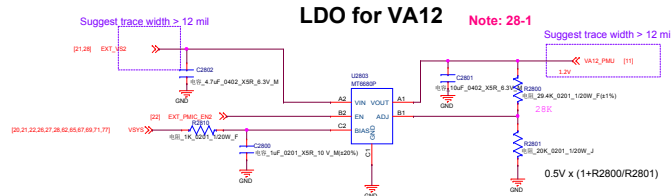


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Note 26-1: For better ESD or surge performance we need choose suitable device for system protection. Please refer to [Surge device selection guide V2.0] provide by MTK.



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Title 27_POWER_SubPMIC-HV powers		
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Schematic design notice of "28_POWER_ThirdParty-Power"

Note 28-1: VA12 Layout placement please close to AP

Note 28-2: VS2 Buck Layout placement please close to PMIC MT6357

Note 28-3: VCN33 LDO Layout placement please close to MT6631

Note 28-4: MT6691ZXP/A and MT6691OOP/A Buck Layout Placement please close to LPDDR4X

Note 28-5: U2810 LDO Layout Placement Please close to LPDDR4X VDD1 power ball

Note 28-6: For EMI_VDDQ_FB and EMI_VDDQ_FB, please follow MMD rule

华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 28_POWER_ThirdParty_Powers

Size D Project F9_MB_V1

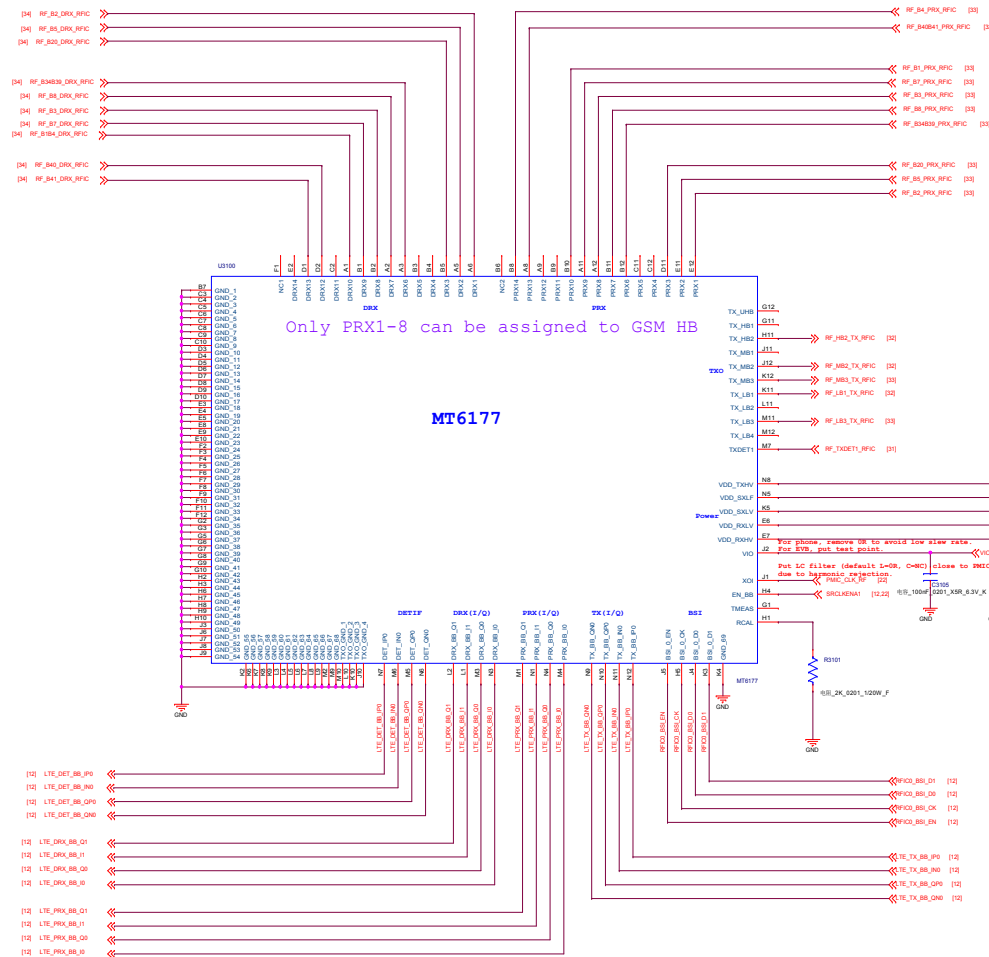
Rev v1

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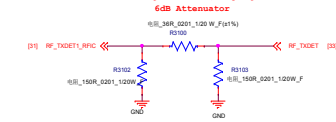
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Attenuator for Detector

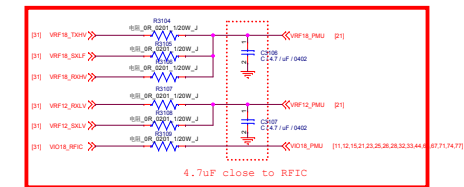
Transceiver



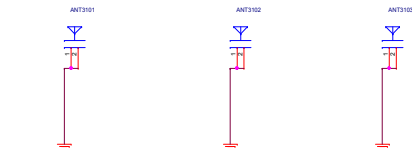
The linear detection range is 0-30dBm, coupling factor is 22-27dB.



Power For MT6177



GND

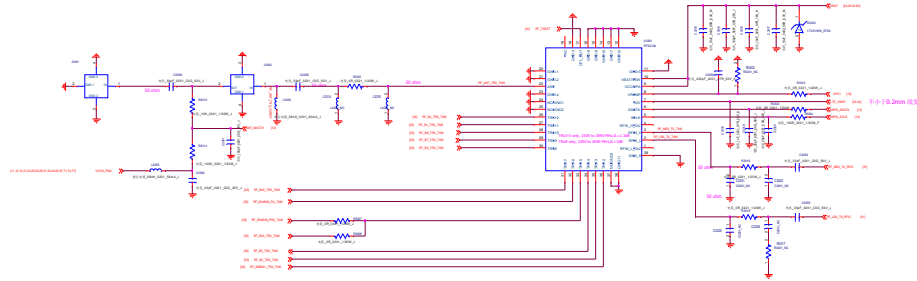


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 31_RF_MT6177_Pin_Out		
Size D	Project F9_MB_V1	Rev v1
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MAIN_ANT

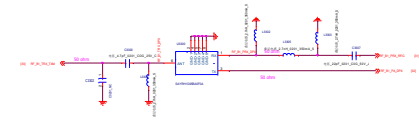
TXM_GSM

0.000000WAT 0.000000WAT 0.000000WAT

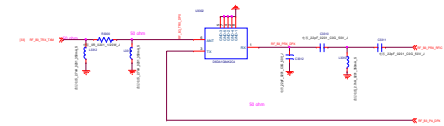


TRX_MB

B1TRX



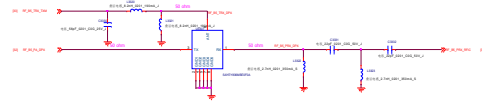
B3 TRX



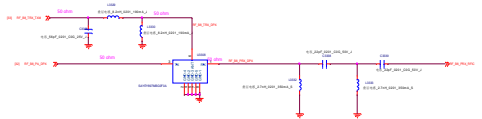
TRX_LB

TRX_LB

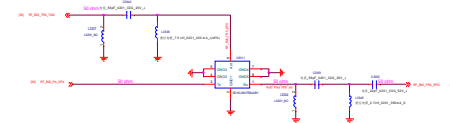
B5 TRX



B8 TRX



B20 TRX

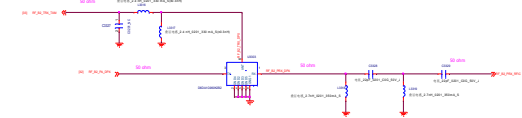


TRX_HB

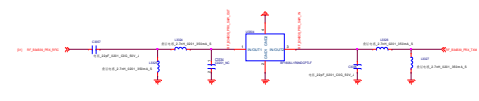
B7 TRX



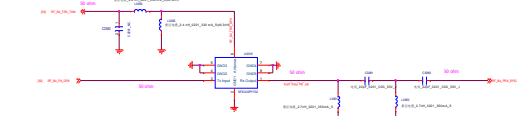
B2 TRX



B34 B39 TRX



B4 TRX



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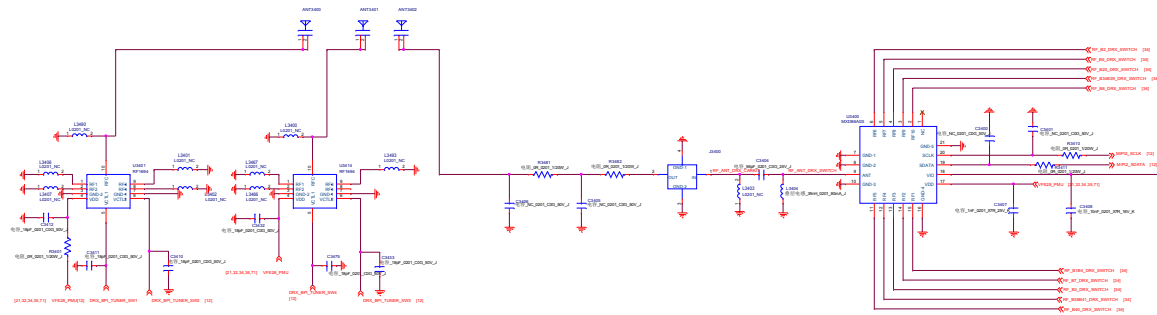
Title 33_RF_MT6177_RF_PRX

Size Project F9_MB_V1
Custom

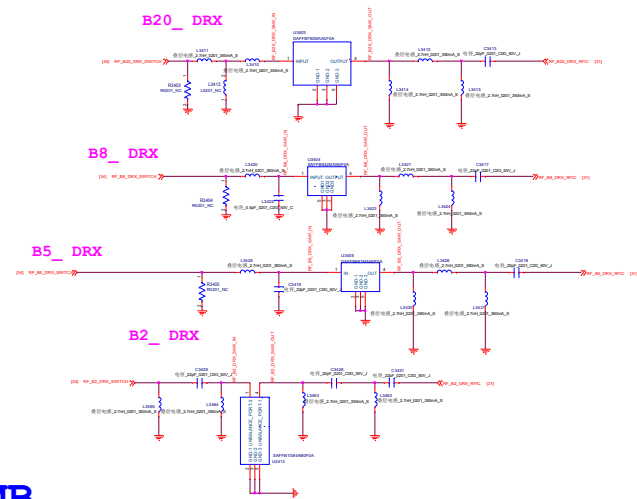
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V1

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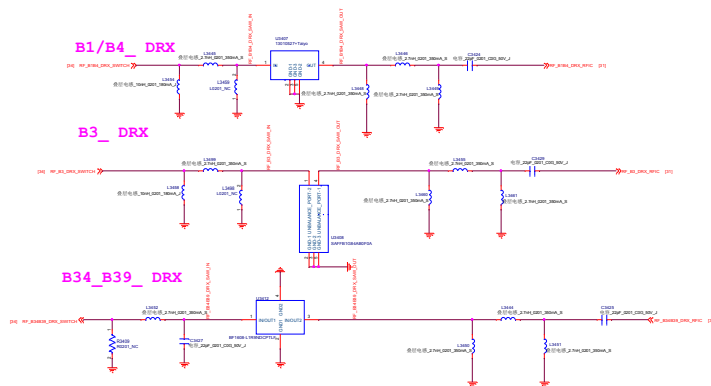
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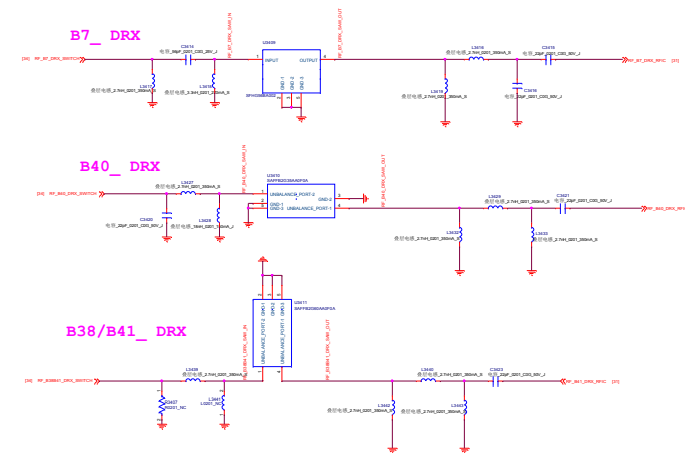
DIV_LB



DIV_MB

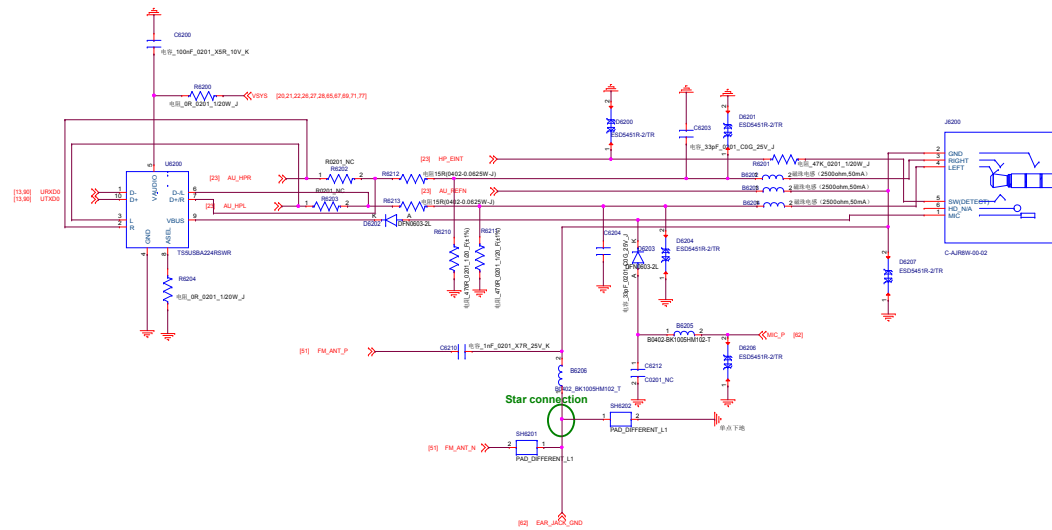


DIV_HB

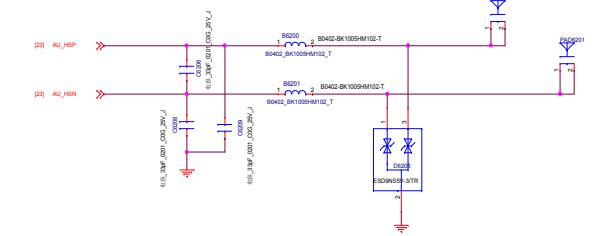


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 34_RF_MT6177_RF_DRX		
Size E	Project F9_MB_V1	Rev v1
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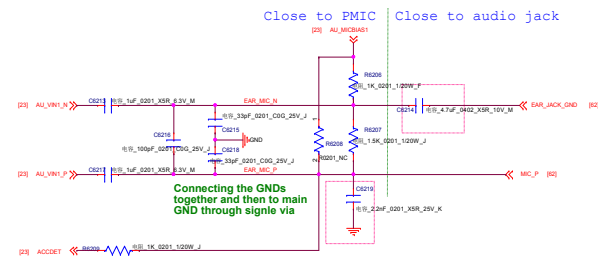
AUDIO JACK



EAR



Earphone Microphone



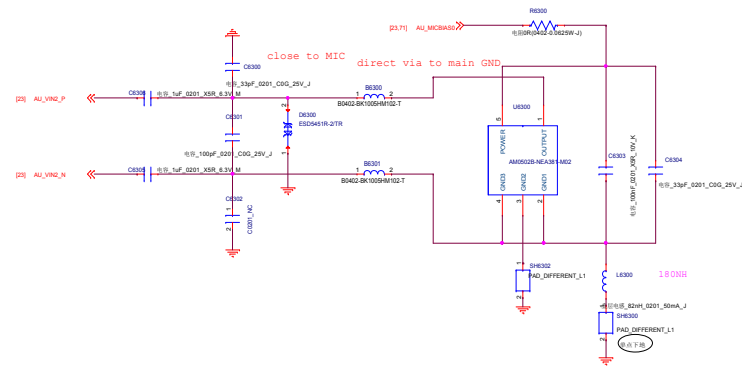
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title	62_PERI_AUDIO_IO
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Size D	Project F9_MB_V1	Rev v1
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MIC -- Sub

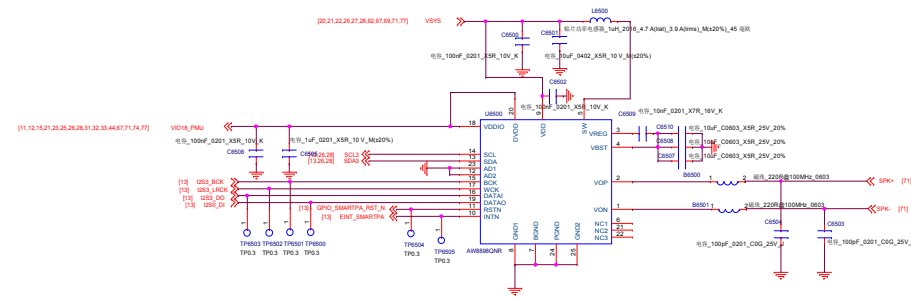


MIC -- Main



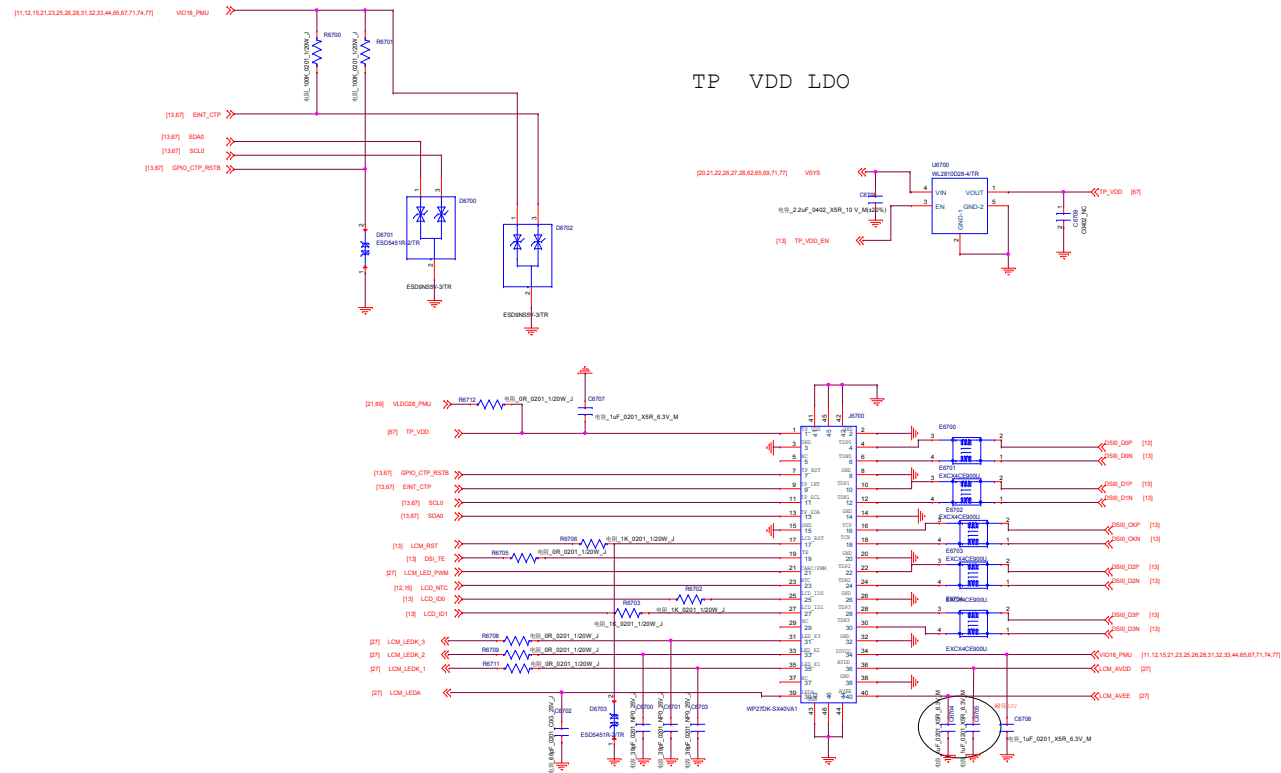
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 63_PERI_AUDIO_IO2(2 AMIC)		
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AUDIO PA



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Title 65_PERI_SPEAKER_AMP		
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Main LCM+CTP



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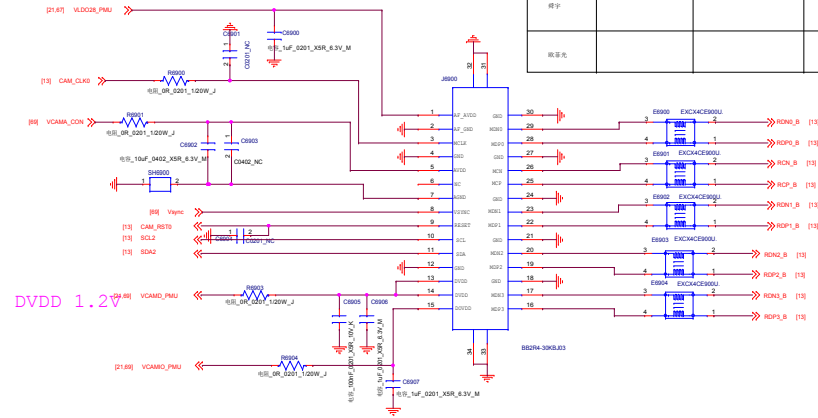
Title	67_PERI_LCD_CTP_FP
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Size D	Project F9_MB_V1
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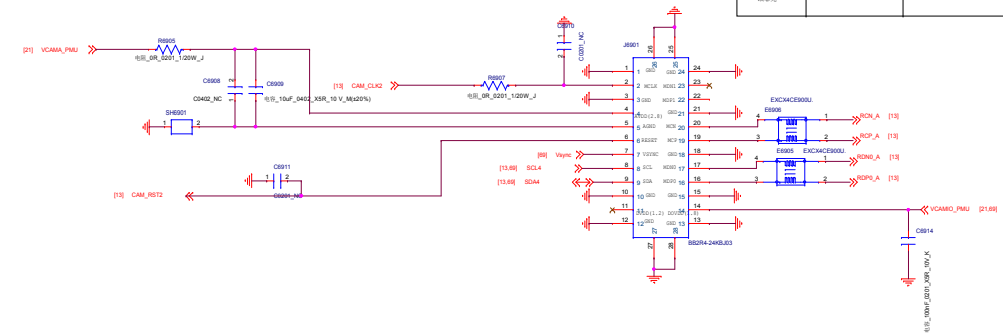
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Main MONO Camera 12M



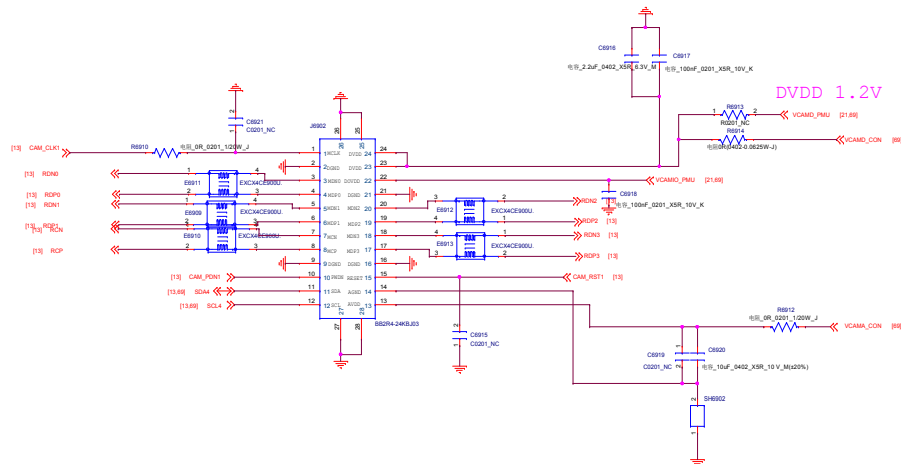
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型号			
欧菲光			

AUX Camera 2M



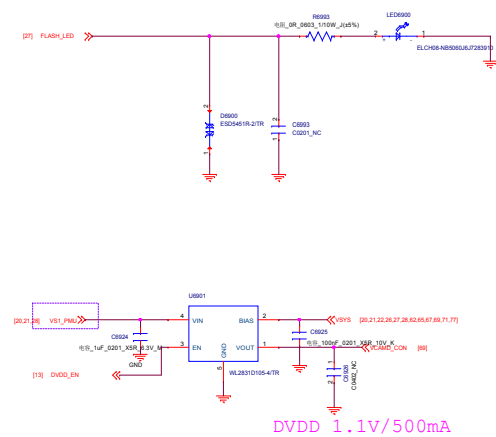
厂家	12C HEAD	12C WASTE	ID
型号			
数量			

Front Camera - 5M

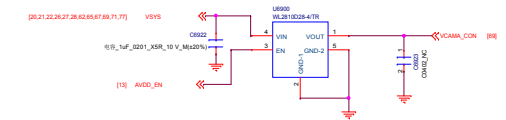


厂家	12C HEAD	12C HEEL	12
丘状磨			
磨面壳			

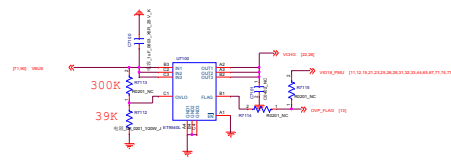
Rear Camera Flash



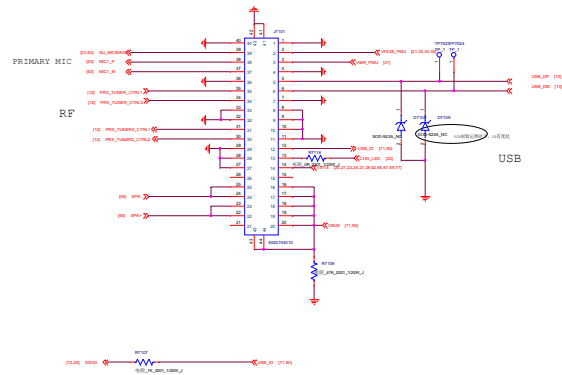
AVDD LDO



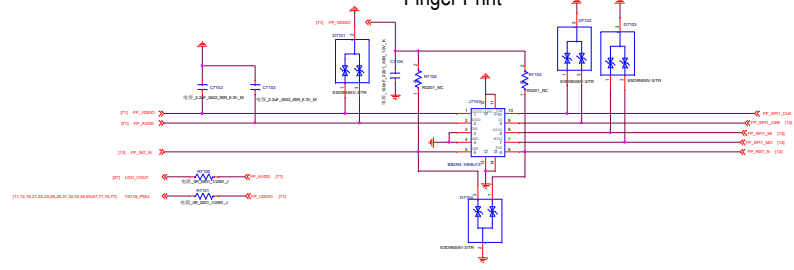
OVP 10.5V



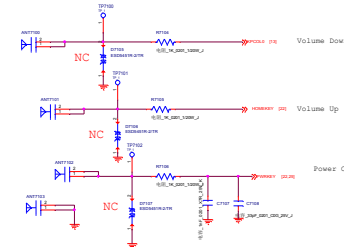
BTB Connector



Finger Print

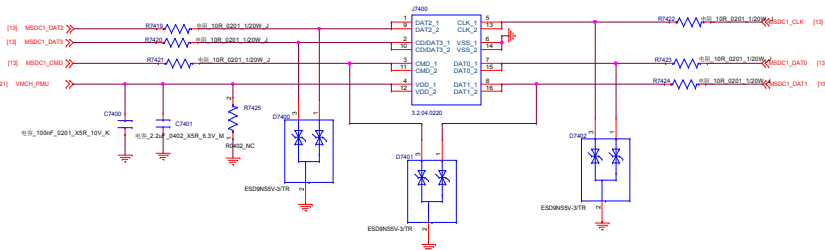


KEY BTB

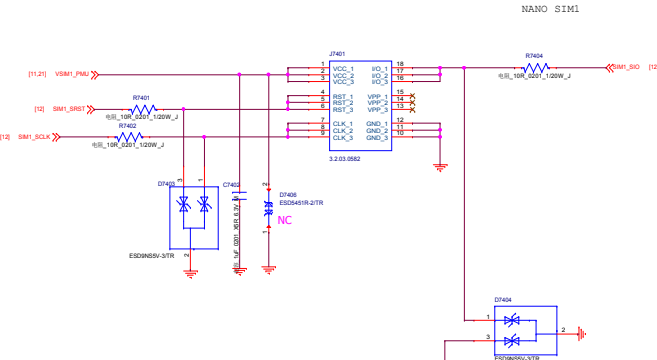


Signal	Description
KPCOLO	Volume Down
HOMEKEY	Volume Up
POWERKEY	POWER_ON

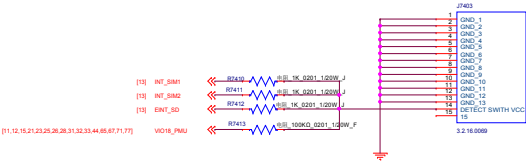
SDC2



SIM1

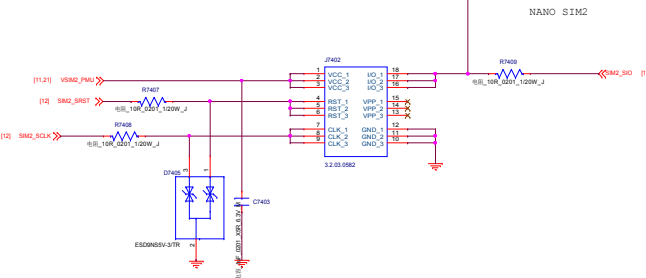


SD_SIM_DET



EINT DEAFULT LOW
plug in int H
plug OUT int L

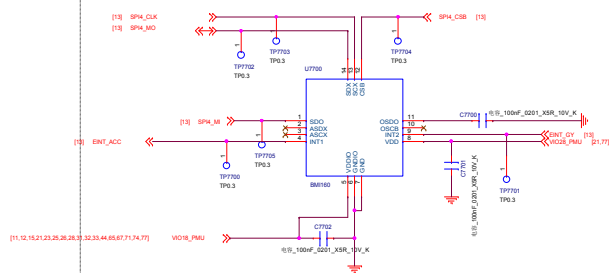
SIM2



Schematic design notice of "46_MEMORY_SD Card" page.
Note 46-1: For better ESD performance, please select suitable components for system protection.

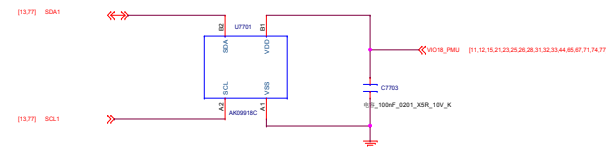
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 74_PERI_SIM&SD Card		
Size D	Project F9_MB_V1	Rev v1
Date: Friday, October 19, 2018		Sheet 74 of 99

Gyro+A sensor



厂家	ISO ADDRESS
000120	0x68

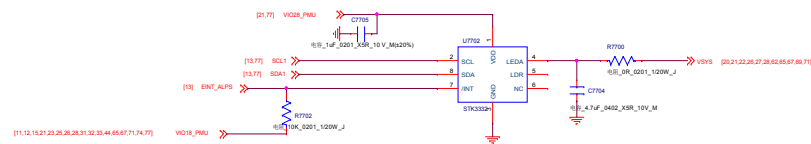
E-compass



厂家	12C AZORESS
AK0991BC	0ch

ALS PS

厂家	INC ADDRESS

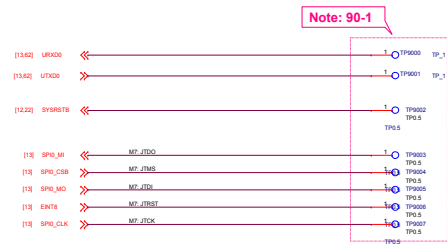


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Title	77_PERI_SENSORS_MEMs_ALS/PS
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Size D	Project F9_MB_V1	Rev v1
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屏蔽罩



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Title 90_DEBUG_IO		
Size D	Project F9_MB_V1	Rev V1
Date: Friday, October 19, 2018		Sheet 90 of 99